Aditya Asutosh Mishra

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Summary

Aspiring Machine Learning Engineer with a strong foundation in computer vision, deep learning, and full-stack development, backed by hands-on project and hackathon experience. Proficient in building and evaluating ML pipelines using PyTorch, TensorFlow, and OpenCV, with backend development experience in Flask and FastAPI. Seeking to apply this foundation to research-driven and defence-oriented ML roles.

Education

Vellore Institute of Technology **Expected: July 2028**
Bachelor of Technology in Information Technology *Vellore, India*

- CGPA: 8.73/10.0

Technical Skills

Programming Languages: C, C++, Java, Python, JavaScript
Web Development & APIs: HTML, CSS, Flask, FastAPI, JSON, REST APIs
Databases: SQLite, PostgreSQL, SQL
Machine Learning & AI: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, Convolutional Neural Networks (CNN), Natural Language Processing (NLP), Deep Learning, Neural Networks, Computer Vision
Data Processing: Feature Engineering, Data Preprocessing, Data Pipelines, Matplotlib, GeoPandas
Tools & Practices: Git, GitHub, Version Control, Jupyter Notebook, Linux

Projects

Deepfake Classification | *Python, TensorFlow, CNN, OpenCV, Scikit-learn* **College Hackathon**

- Engineered a deep learning pipeline to classify real vs. deepfake images using Convolutional Neural Networks (CNNs), trained on a hackathon-provided labeled dataset of real and AI-generated faces.
- Applied data augmentation (rotation, flipping, brightness adjustment) via OpenCV and TensorFlow data pipelines to improve model generalization and reduce overfitting.
- Designed a multi-layer CNN architecture with iterative hyperparameter tuning (learning rate, batch size, layer depth) to optimize validation performance.
- Evaluated model performance using accuracy, precision, recall, and F1-score as core classification metrics.
- GitHub: github.com/adityaDev-2005/model_arena_hack

Satellite Image Generator | *Python, GeoPandas, Shapely, Contextily, Matplotlib*

- Engineered a Python-based geospatial remote sensing tool to generate high-resolution satellite images from location names or GPS coordinates, eliminating the need for specialized GIS software.
- Built end-to-end data pipelines using GeoPandas and Shapely for coordinate transformation, bounding-box computation, and vector overlay rendering on raster tile maps.
- Integrated Contextily for dynamic web map tile fetching and Matplotlib for image export, supporting customizable zoom levels, resolutions, and multiple tile providers.
- Identified extension potential for obstacle detection on generated imagery, applicable to autonomous vehicle and drone navigation pipelines.
- GitHub: github.com/ToxicKoder/Satellite-image-generator

Drowsiness Detection System | *Python, OpenCV, dlib, Machine Learning*

- Architected a real-time computer vision system for driver drowsiness detection using facial landmark detection and the Eye Aspect Ratio (EAR) algorithm to classify eye closure states.
- Achieved 84.56% overall accuracy with 1.0 precision (zero false positives) on evaluation data; identified low recall (0.24) as a threshold-tuning tradeoff for future iteration.
- Deployed a threshold-based alert mechanism triggering audible warnings on prolonged eye closure, prioritizing false-alarm avoidance over detection sensitivity, simulating a driver safety assist system.
- GitHub: github.com/ToxicKoder/Drowsiness-Detection-System

Achievements

UCO Bank Hackathon — Top 20 Finalist
IIT Kharagpur

- Selected among the Top 20 finalists at the PSB Hackathon (UCO Bank x IIT Kharagpur), addressing the problem statement on AI voice-cloning fraud in banking.
- Developed **Dhwani-Kavach**, a real-time audio forensics and voice fraud detection system combining synthetic-voice (deepfake audio) detection via spectrogram analysis, voice authentication, threat scoring, and conversational analysis to flag high-risk calls within 10 seconds, while prioritizing privacy and on-premise deployment.